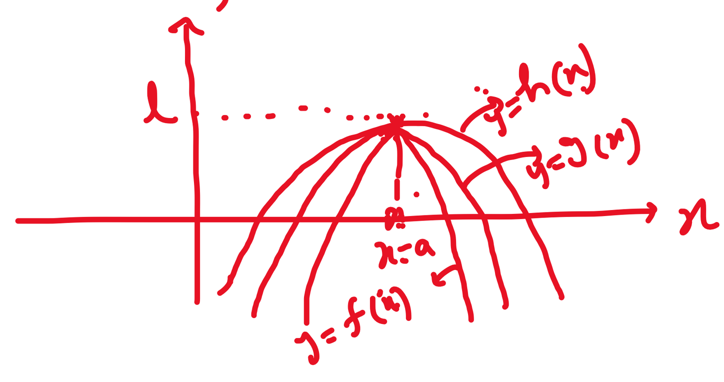
Sandwich theorem / Squeeze theorem

Statement: Let f, g and h be real functions such that $f(x) \le g(x) \le h(x)$ for all x in the common domain of definition. For some real number a, if

$\lim_{x \to a} f(x) = l = \lim_{x \to a} h(x)$, then $\lim_{x \to a} g(x) = l$



Ex: Find $\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right)$ by sandwich theorem.

$\frac{0}{0} / \frac{\infty}{\infty}$

$\lim_{x \to 0} \frac{\sin(1/x)}{(1/x)}$

Sol^n: We know,

$-1 \le \sin x \le 1$

$\Rightarrow -1 \le \sin\left(\frac{1}{x}\right) \le 1$

$\Rightarrow -x^2 \le x^2 \sin\left(\frac{1}{x}\right) \le x^2$

$\lim_{x \to 0} \sin\left(\frac{1}{x}\right) = c$

$\lim_{x \to 0} \frac{\sin x}{x} = 1$

Now, $\lim_{x \to 0} -x^2 = 0$ and $\lim_{x \to 0} x^2 = 0$

By the squeeze/ Sandwich theorem, we get

$\lim_{x \to 0} x^2 \sin(1/x) = 0$

L' Hospital's rule

$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 0^0, 1^∞, ∞^0

indeterminate form

$\left(\frac{0}{0}\right) / \left(\frac{\infty}{\infty}\right) \longrightarrow$ L' Hospital's rule

$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0} / \frac{\infty}{\infty}$

$\lim_{x \to a} \frac{f'(x)}{g'(x)}$

$\frac{df}{dx} = f'(x)$

$\frac{dg}{dx} = g'(x)$

Ex: Find $\lim_{x \to 0} \frac{e^x + e^{-x} - 2}{x^2}$ using L' Hospital's rule.

Sol^n: $\lim_{x \to 0} \frac{e^x + e^{-x} - 2}{x^2}$ $\left[\frac{1+1-2}{0^2} = \frac{0}{0} \text{ form}\right]$

$\frac{d}{dx}(e^{-x}) = -e^{-x}$

$\frac{d}{dx}(e^x) = e^x$

$\frac{d}{dx}(x^n) = nx^{n-1}$

$\frac{d}{dx}(c) = 0$

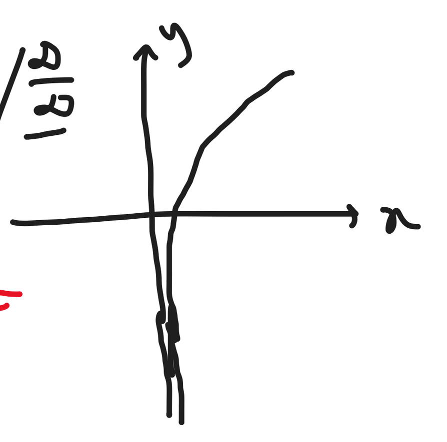
Applying the L' Hospital's rule,

$\lim_{x \to 0} \frac{e^x - e^{-x}}{2x}$ $\left[\frac{1-1}{2.0} = \frac{0}{0} \text{ form}\right]$

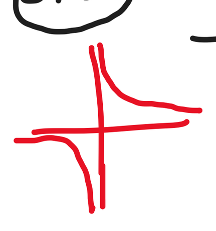
Applying the L' Hospital's rule

$\lim_{x \to 0} \frac{e^x + e^{-x}}{2} = \frac{1+1}{2} = 1$. Ans

Ex: Find $\lim_{x \to 0} x \ln x$ $\quad 0 \times (-\infty) = -0 \times \infty$ $\quad \frac{0}{0} / \frac{\infty}{\infty}$



Sol^n: $\lim_{x \to 0} x \ln x$ $\quad \left[\frac{g(x)}{h(x)}\right]$



$= \lim_{x \to 0} \frac{\ln x}{(1/x) = x^{-1}}$ $\left[-\frac{\infty}{\infty}\right]$ $\quad \frac{1}{0}$ $\quad \frac{x \to}{(1/\ln x) \to}$

$\frac{d}{dx}(\ln x) = \frac{1}{x}$

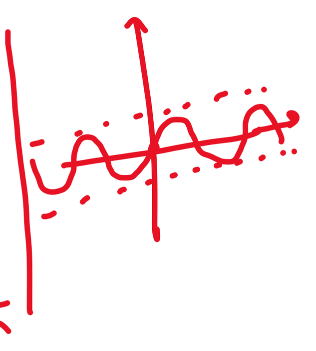
Applying the L' Hospital's rule,

$\lim_{x \to 0} \frac{1/x}{-x^{-2}} = \lim_{x \to 0} -\frac{1}{x} \times x^2 = \lim_{x \to 0} -x = 0$

$\therefore \lim_{x \to 0} x \ln x = 0$. Ans

Ex: Find $\lim_{x \to 0} (\sin x)^{\tan x}$ using L' . . .

$\frac{0}{0} / \frac{\infty}{\infty}$



$\tan x = \frac{\sin x}{\cos x}$

Sol^n: $\lim_{x \to 0} (\sin x)^{\tan x}$ $\quad [0^0 \text{ form}]$ $\quad \ln x^y = y \ln x$

let, $z = \lim_{x \to 0} (\sin x)^{\tan x}$

Taking $\ln$ on both sides, we get

$\ln z = \lim_{x \to 0} \tan x \ln(\sin x)$ $\quad [0 \times \infty \text{ form}]$

$= \lim_{x \to 0} \frac{\ln(\sin x)}{\cot x}$ $\quad \left[\frac{\infty}{\infty} \text{ form}\right]$

$\frac{d}{dx}(\cos x) = -\sin x$

Applying the L' Hospital's rule,

$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

$\frac{d}{dx}(\sin x) = \cos x$

$\lim_{x \to 0} \frac{\frac{1}{\sin x} \times (\cos x)}{-\operatorname{cosec}^2 x}$

$= \lim_{x \to 0} \frac{\cos x}{\cancel{\sin x}}\left(-\cancel{\sin^2} x\right)$ (sin x) $= -\lim_{x \to 0} \cos x \cdot \sin x$

$= -1 \cdot 0 = 0$

$\ln z = 0$

$\Rightarrow z = e^0 = 1$. Ans